Chapter 1

Translation competence in the age of generative AI: Debates, dilemmas, directions

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Until recently, the description and modelling of the competences and skills needed to translate successfully, and the ways in which they develop, have seen a steady evolution and predictable expansion, largely to accommodate technological advances and an increasing awareness of situatedness. However, the impact of neural MT and, now, generative AI (GenAI) has been unprecedented in rapidly transforming the core tasks of translators. Together with a proliferation of creative roles in a diversifying language industry, the volatility indicates a paradigm shift that is beginning to supplant even the once stable epithet "translator". It also questions the efficacy of current descriptions of skills and confronts educators with dilemmas of balancing specialisation and generalisation, routinisation and adaptivity, core and transferable skills. This chapter considers relevant aspects of modelling competences and their development and examines related debates and dilemmas engaging educators and employers in the current and foreseen language-industry climate. It outlines directions for training in the age of translating with(out) GenAI, proposing an approach that, alongside core textual, interlingual translation and digital skills, combines transferable skills with human-machine/human-agent interaction (HMI/HAI) competence.

1 Introduction

1.1 General

What elements constitute translation competence (TC) and how they can be developed have represented pivotal issues in applied Translation Studies since its

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inception (**Holmes1988**). They continue to trigger considerable debate amongst researchers, educators and practitioners, and they present educators and their organisations with dilemmas about how and where to target resources. In this chapter, we explore some debates and dilemmas triggered by changes to professional and educational practices brought about by generative artificial intelligence (GenAI). We also outline potential directions for human agents in the age of translating with and without GenAI. In our view, this necessitates an approach combining transferable skills with human-machine/human-agent interaction (HMI/HAI) competence, alongside "old-school" translation skills. Before proceeding to the debates, dilemmas and directions, however, we should define, describe and demarcate what exactly we mean by professional interlingual TC in the age of GenAI.

1.2 Definitions, descriptions and demarcation

The word interlingual is used in Translation Studies and the language industry to designate a particular type of language mediation activity, namely that which takes place between natural languages and the particular cultures they represent. Alongside interpreting, translation represents a prototypical form of professional interlingual language mediation. In a language industry characterised by rapidly diversifying job titles and tasks (**Bond2018**; **Slator2020**: 11-17), the prototypical conceptualisation of what so-called (professional) translators do, i.e. translate content written in a source-language (SL) document into a target-language (TL) document, appears increasingly outmoded. Language professionals adaptively engage in a whole range of service provision as the lines between core and adjacent services blur (**Angelone2023**; **AngeloneEtAl2024**: 3-5).

The major driver behind this state of affairs has been the technologisation of the industry as language service providers (LSPs) operate more and more downstream and upstream of previously core translation, localisation and interpreting activities. Indeed, the industry itself prefers the generic term linguist, and with good reason. **Since2016**, the higher accuracy, fluency and adequacy of neural machine translation (NMT) engines have led to more extensive automation of the translation process. Linguists have thus seen more deployment downstream of language transfer (or conversion) per se in post-editing (PE), output evaluation and quality assurance tasks. Most recently, the introduction of automated MT quality estimation (QE) to determine how much human PE is required is itself replacing more costly human MT quality evaluation (**Slator2024**). Upstream of transfer proper, linguists have been expanding their repertoire to perform tasks that include linguistic consultancy, MT pre-editing and multilingual content creation, for example in marketing, PR and corporate communications (often referred to as transcreation). Nevertheless, we shall retain the term translation, both in line with the title of the present volume and in recognition of do Carmo and **Moorkens2021**' (**Moorkens2021**) convincing argument that the ultimate ethos of translation is to help communication flow between humans, regardless of the new technologies deployed and of the roles and activities these necessitate.

GenAI has now entered the technological mix, and its applications clearly go far beyond interlingual mediation. Opportunities and risks have been identified in higher education in general (**Atlas2023**; **GimpelEtAl2023**; **OECD2023**), where the potential of GenAI as an educational tool extends to teachers and students alike. Properly used, it can support course design, materials development, assessment, text and image production, coding, critical thinking and individualised learning. Improperly used, it can facilitate academic dishonesty, increase technological dependency, atrophy human skills and agency, undermine data protection and intellectual property rights, reinforce discrimination (due to data bias) and increase inequality (due to unequal access).

Within the narrower confines of interlingual translation, GenAI differs decisively from AI-based language technology such as NMT in being capable of generating multilingual text, images and other content in response to prompts, including the prompt to translate. These prompts elicit statistically probable output generated by deep-learning models able to process and generate natural languages, known as large language models (LLMs). The prompt-response cycle can be iterative, allowing users to ask for more information or to provide more detailed prompts should the results require improvement (**GimpelEtAl2023**: 22). Adequately structuring the instructions that prompt a GenAI model – also known as prompt engineering – thus becomes central to its effective use.

The skills and competences necessary to embrace GenAI in the education and practice of professional translators thus differ in major respects from those needed to master CAT tools and other translation technologies. In the present chapter, we apply the definitions of skills and competences from the flagship European Master's in Translation Competence Framework (**EMT2022**), namely the "ability to apply knowledge and use know-how to complete tasks and solve problems" (skills) and the "proven ability to use knowledge, skills and personal, social and/or methodological abilities, in work or study situations and in professional and personal development" (competence).

It is indisputable that such skills and competences are essential to a language industry already transitioning from NMT to GenAI. Unbabel co-founder and

CEO João Graça points out that LLMs are attracting more research and development, and can handle more complex tasks than NMT, such as automatic PE, source correction and cultural adaptation.[1] Lionbridge, one of the world's leading LSPs, claims GenAI is "taking automated translation and localisation to new heights".[2] **Slator2024** reports that translation management systems (TMS) providers are already integrating GPT models to leverage LLMs' capacity to rephrase, summarise and suggest new translations. LLMs can be used to incorporate QE into workflows and gauge PE effort, do PE or even provide linguistic insights throughout the translation workflow. Slator's results also show two-thirds of MT providers offering fine-tuned LLMs and over 80% offering MT-LLM hybrid solutions. LSPs have the potential to become major providers and enablers of multilingual, multimodal GenAI content through content creation, validation, localisation, transmission and management. In keeping with the broadening portfolios of LSPs, multilingual text generation is currently "the most in-demand language AI application, after machine translation" (**Slator2024**). Fine-tuning and prompting LLMs to create and translate/localise content are emerging activities for language professionals to complement longer-established PE and quality assurance roles.

Competent – or ideally expert – leverage of GenAI is a must for all those aiming to work in the language industry. But which skills and competences does and will this require? And in a dawning age of (language) work increasingly dominated by AI, what value can human intelligence and skills bring to bear? It is time to consider the debates around professional interlingual translation and the dilemmas with which educators are faced.

2 Debates and dilemmas

2.1 Agency

Asscher2023 has recently raised the issue of whether MT can actually be regarded as a proper point of interest in Translation Studies from a definitional perspective. He concludes that it is indeed compatible with both prescriptive, equivalence-oriented definitions of what translation is as well as descriptive definitions based largely on how translations are received. But he also points out that the perceived threat of AI to the social and professional status of translators, and of those who educate them (see also **ELIS2024**: 24-25), may well redraw

[1]Slatorpod #216, 12 **July2024** (https://www.youtube.com/watch?v=6smTEp3CXwQ)

[2]https://www.lionbridge.com/generative-ai/

definitional boundaries and distinguish between MT and human translation on the basis of the perceived creative and moral authenticity of conscious agency (**Asscher2023**).

Asscher's argument draws on current positions on AI from the broader humanities and social sciences, such as moral philosophy (e.g. **SebastiánRudy-Hiller2021**). However, there is evidence of similar preoccupations in education policy and science, with emphasis falling on the importance of human agency, including teacher agency, and associated transferable skills – responsible action, critical thinking, systems thinking, logical reasoning, cultural agility, problem solving and emotion regulation (**GimpelEtAl2023**; **OECD2023**).

Within the narrower confines of Translation Studies, it appears that agency is already emerging as the fulcrum around which the value of professional human translation can be measured. For example, **Vieira2020** presents PE as a spectrum ranging from an MT-centred process typified by automatic PE to human-centred PE where humans have full control of output. Human agency moves back and forth along the spectrum as client, commission and employer demands require. The human-centred pole is more palatable to professional translators (**Vieira2020**), whose resistance to PE seems to derive from anxiety over a perceived loss of agency (**CadwellEtAl2018**; **Sakamoto2019**). In the discourse surrounding PE, agency has thus become a touchstone of professionalism and professional self-concept.

Similarly, the translation educators in Rico and Gonzalez Pastor's (2022: 188) study of attitudes to teaching MT make much of the "human factor". Though the data are unclear on precisely where they believe that agency resides, the participants place the translator at the centre of translation production. Interestingly, this prompts the researchers to posit that "MT has overcome its condition as a tool", and to claim that teaching MT should adopt a holistic approach "beyond an instrumentalist agenda that concentrates on the technical properties of the technologies" (Rico and Gonzalez **Pastor2022**: 190). The implication is that the paradigmatic human agency of the tool-user is being subverted by a more discerning and subtle appreciation of the rapidly evolving interactivity between translation technologies and their users.

The changed nature of that relationship is reflected in the metaphors used to describe GenAI systems: ChatGPT and similar systems are referred to as "conversational agents" (Moorkens and Guerberof **Arenas2024**: 75), and educational sources describe them as a "language partner", "writing partner", "writing collaborator", "co-partner to formulate text", "learning partner", "intellectual sparring partner", or "partner to generate codes" (e.g. **Atlas2023**: ii: 64; **GimpelEtAl2023**:

19-24). Business consultants and IT case researchers alike refer to human-GenAI interaction as "co-creation" (e.g. **EapenEtAl2023**; **NahEtAl2023**: 296).[3]

GenAI has now advanced well beyond the material agents in the "dance of agency" so deftly explored by **Olohan2011** in interactions between translators and TM. This is a question not just of degree (speed, accuracy and adequacy of suggestions, etc.) but also, and most pertinently, of quality (reciprocal interactions and learning). The model pivotal to Olohan's (2011: 344) conceptualisation of agency decisively distinguishes between the capacity of human agents to both resist and accommodate to technologies and of non-human agents to simply resist. But GenAI is patently capable of accommodation in its own right (deep learning). This renders irrelevant the perceived divide between the poles of social and technological determinism that **Olohan2011** observes among translators commenting on TM, and which O'**Brien2024** criticises in her call to overcome the antagonism between technology and translators by focusing on a human-centred AI (HCAI) that amplifies rather than emulates human abilities (cf. **Shneiderman2020**). As Risku and **Windhager2013** point out, any technology or artefact used by translators in their work aligns with the concept of "actants" in actor-network theory (ANT, e.g. **Latour2005**). GenAI, however, quite obviously goes further in fulfilling the conditions of what the human-machine interaction (HMI) sub-field of human-agent interaction (HAI) has called "joint activity" (**BradshawEtAl2011**: 288): "the essence of joint activity is interdependence [...] to produce something that is a genuine joint product".

Together with other effects of (Gen)AI on the processes and practices of professional translation, the agency issue yields multiple potential ramifications for the development and exercise of professional TC. It is to these that the next sections turn.

2.2 Modelling competence

The reciprocity and mutuality of GenAI already go some way towards realising aspects of human-centred augmented translation so engagingly discussed by O'**Brien2024**. She argues for HCAI as a framework for amplifying translators' abilities and empowerment while maintaining human control. We would

[3]The anthropomorphism is more than a conceit. Research on conversational agents used as social companions shows that human-AI friendship is similar to human-human relationships and contributes to social health (**BrandtzaegEtAl2022**; **ChaturvediEtAl2023**; **GuingrichGraziano2023**). Virtual agents capable of reciprocal adaptive behaviour are being tested for use in cognitive behaviour therapy and social skills training (**WooEtAl2024**).

claim that GenAI represents a collaborative technology capable of complementing human agency and supporting empowerment to a degree unseen in dedicated (augmented) translation technologies to date. Even before ChatGPT launched the GenAI era late in 2022, **FügenerEtAl2021** could confidently assert that "human performance can improve individually by receiving AI advice [...] due to complementary knowledge". Although their results also showed "significant downsides" when AI advice is provided as a "one-size-fits-all" solution in group decision-making, the individualisation made possible by appropriate prompt engineering would suggest that this is no longer an issue.

As in all complementary partnerships, the actants (to use the ANT term) must bring their own strengths to the table. Competent translators will for their part still need the transferable human skills that enable them to regulate cognitive and affective performance without AI support, the experience and knowledge on which they are predicated, and the competences necessary to deploy them within the sociotechnical environments where they work. These will have to be aligned with the new key skill of interacting with GenAI by prompting and reprompting LLMs,[4] based on knowledge and experience of how GenAI works. We further propose that this skill could be embedded in a broader area of HMI/HAI competence (see below) which takes due account of GenAI's technological agency.[5]

It is here that current TC models reveal inadequacies. Although they give due weight to the transferable personal and interpersonal skills that facilitate human agency, they have perpetuated an instrumentalist teaching agenda (Rico and Gonzalez **Pastor2022**: 190) by focussing almost exclusively on aspects of CAT tool use (including MT) that assume no technological agency. For example, in the case of the EMT competence frameworks, the only hint of possible reciprocal effects between the technological environment and human performance has been the inclusion of an organisational and physical ergonomic descriptor as an aspect of personal competence (**EMT2022**). Tellingly, it makes no mention of ergonomic factors of technology design and use that play into cognitive performance, despite empirical evidence that they can be major enhancers, distractors or stressors (**Ehrensberger-DowEtAl2016**). Otherwise, the EMT technology competence category focuses on non-interactive aspects of technology use: critically assessing, adapting to and effectively deploying available resources, including appropriate use of MT, file management and TMS, and demonstrating data literacy.

[4]See Chapter 5 by M. Yamada in the present volume.

[5]This is not to say that there is equivalence between conscious human agency and AI agency. However, they share features that suggest "family resemblances" in the sense developed by Wittgenstein (e.g. 1958: 32).

Interactivity is visible only in interpersonal relations: checking, reviewing, revising and evaluating the work of others, working in multicultural and multilingual teams, interacting with clients and networking with language professionals and LSPs (**EMT2022**).

The older but no less influential PACTE Group's competence model (Hurtado **Albir2017**: 35-41) is similar in this respect. It includes technology under what is called instrumental sub-competence, "related to the use of documentation resources and information and communication technologies applied to translation" (Hurtado **Albir2017**: 40). The descriptor again demonstrates how competence models have fuelled the instrumentalist agenda criticised above.

Nevertheless, the PACTE model has been a major source for others, who have supplemented it with additional components where appropriate. Those additions reflect a growing awareness that translation is far less a discrete cognitive act performed solely by the mind of an individual than an event of embodied cognition performed by complex systems involving humans, their social, technical and physical environments together with their cultural artefacts (**Risku2010**). For example, an interpersonal component is added to the most recent PACTE-inspired model: Prieto **Ramos2024a**'s (**Ramos2024a**) adaptation of his previous model of legal TC (Prieto **Ramos2011**) to institutional translation, ostensibly designed for the AI age. The model's "instrumental competence" category builds on the PACTE model and specifies the use of reliable resources for information mining, of computer tools for translation and revision tasks, and of MT (Prieto **Ramos2024b**: 155). But it completely ignores the implications of human interactivity with agentic AI systems.

Despite some overt (and acknowledged) borrowings, Prieto **Ramos2024b** is highly critical of the latest EMT framework (2022) for failing to redress what he sees as a downgrading of thematic competence (i.e. domain specialisation, e.g. in law) in its previous iteration (**EMT2017**). It is hard to argue against him, since thematic competence figures high on the list of professional expectations uncovered by his own research and that of others, especially with respect to legal and institutional legislative translation (**EsfandiariEtAl2019**; **Lafeber2023**; Prieto **RamosGuzman2023**). The research receives equal credence from emic language industry heuristics. Describing **Slator2022**'s (**Slator2022**) "expert in the loop" model, for example, industry commentator Florian Faes remarks that only language professionals with high levels of domain and language expertise, combined with appropriate technological and prompt engineering competence, will provide the added human value needed to work with (Gen)AI and that such experts will stay in high demand (**FaesMassey2024**: 27-29).

In other respects, however, the EMT framework (2022: 10) falls in with the broad consensus, already articulated above, that transferable personal and interpersonal skills (called generic or soft skills) are central to employability and to the adaptability needed in today's work environment. The point is echoed by a recent PE model (Nitzke and Hansen-Schirra, 2021: 69-79), visualised as a house, where the basic bilingual, extra-linguistic and information research competences that professional translators possess form the foundations. The roof is made of soft skills such as risk assessment, service competences, self-efficacy, a professional self-concept, an ethical attitude, concentration, stress-resistance, logical reasoning, analytical thinking and an affinity with technology. The model is one of several developed for tasks like revision and PE (**RobertEtAl2023**) which were once considered integral aspects of TC. These reflect the growing diversification of language-industry tasks and the concomitant need for language professionals to have transferable skills enabling them to adapt to new activities.

2.3 Developing competence

Industry diversification has prompted **Angelone2023** to call for fostering adaptive (as opposed to routinised) expertise across translator training curricula by consistently exposing learners to challenging situations where they need to apply their knowledge flexibly – itself a transferable skill set. But what balance should be struck between routine and adaptivity, between core and transferable skills? These are two dilemmas confronting translation teachers and their institutions with limited material and temporal resources at their disposal. Related to them is a third, namely employability tensions between domain or task specialisation (Prieto **Ramos2024a**) and a generalist approach more congruent with adaptivity training (**EMT2022**). In all these cases, choices have to be made in specific educational contexts as to the most likely prospects for graduate employment, with curriculum developers nimble enough to adapt quickly to market changes.

The most widespread issue, however, is not unique to translator education. In the context of GenAI use in society in general, **TeubnerEtAl2023** refer to the "educators' dilemma" – between banning GenAI use, as many translation teachers might have been inclined to do with freely available, high quality NMT post-2016,[6] and educating students about how to use it well. Their prediction (**TeubnerEtAl2023**: 97) about "a shift of focus away from the act of writing itself towards the actual content and the ideas being communicated" is very pertinent to the context of translation training. They suggest that this could lead to more

[6] But see **Mellinger2017** for a well-timed argument for not doing so.

importance being placed on reading and evaluating options, recalling **Pym2003**'s (**Pym2003**) minimalist proposal to describe TC.

Viewed through the prism of HMI/HAI, GenAI can be regarded as an interactive, agentic partner in translation production and learning processes. Conceptualising the role of GenAI in this way seamlessly integrates into social constructivist and emergentist experiential learning (**Kiraly2000**, **Kiraly2013**). Here, learning is understood as a collaborative process of work on authentic or simulated translation projects, supplemented with appropriate scaffolding. Teachers essentially partner with their students as learners in proactively seeking knowledge (**Kiraly2013**). Various qualitative studies have validated the approach, with reported outcomes including increased student responsibility, learner autonomy, critical reflection, self-regulation, motivation and self-efficacy (**KiralyMassey2019**) – those very transferable skills that educators see as a necessary complement to AI and which O'**Brien2024** would want AI to support.

Alongside such transferable skills, the language industry continues to expect graduates with skills conventionally associated with translation, such as high reading proficiency in the SL and writing/editing proficiency in the TL. The expert in the loop model (**Slator2022**) assumes that the translator in that position also has a solid understanding of the target audience, awareness of cultural differences, sociolinguistic competence and media competence. These "old school" skills have many parallels to what the **EMT2022** subsumes under language, culture and translation competence. In the next section, we discuss some possible directions for translator training that include GenAI and that had not been foreseen in the competence models proposed to date.

3 Directions: Skills and competences

3.1 Translating with(out) GenAI

Before discussing the skills and competences for translation in an age of GenAI, it makes sense to reflect briefly on what the activity of producing professional translations involves. This can be done by translating from scratch,[7] working with CAT tools or using MT or GenAI to produce a whole target text (TT).

[7] Simply referred to as translation before the widespread introduction of language technology (**ChristensenEtAl2017**).

3.1.1 3.1.1 Translating from scratch

Translating from scratch refers to translators coming up with their own version of a TT on the basis of a source text (ST) without having matches or suggestions presented to them automatically by a CAT tool or an MT/GenAI engine, respectively. To do so, they rely on their SL competence to understand the ST, their translation skills to transfer the meaning and message into the TL, and their TL competence to produce a TT appropriate to the needs of the target audience. Appropriateness includes being aware of and observing genre conventions, language register, terminology requirements and client style guidelines. Translation skills include following a translation brief, assessing the target audience needs, relying on internal resources and common sense, acquiring domain knowledge as needed, accessing terminological resources and/or parallel texts, recognising cultural references and/or metaphors, searching for and adapting solutions, etc.

Translating from scratch does not exclude the option of accessing digital resources such as online dictionaries, concordances in CAT tools, MT engines and even GenAI, but the translator is in full control of when and how those resources are used for which parts of the TT. While drafting the TT and almost always also after producing a complete draft, professional translation involves carefully rereading the TT and comparing it with the ST in a bilingual review to ensure completeness, accuracy and appropriateness.

3.1.2 3.1.2 Producing translations with a CAT tool

Working with a CAT tool to produce professional translations differs from translating from scratch in that the ST is segmented by the tool, and previously translated segments that are perfect matches or similar to a minimum match level are offered for the translator's approval or revision. Sometimes even perfect matches are not appropriate for the particular TT being produced (because of terminology, register, client style guidelines, etc.), so the translator has the choice of editing them or deleting them to translate from scratch. Many CAT tools now also have the option of accessing MT suggestions for segments or parts of segments that do not have good matches in the TM, either automatically or at the command of the translator.[8]

Usually much less typing is involved than in translating from scratch, but otherwise using a CAT tool to produce a TT requires a very similar set of skills. However, there are additional challenges in using a CAT tool to produce a TT.

[8] See **Kappus2024** for an overview of the development and convergence of translation technology.

One is the focus on text segments (e.g. sentences or parts of sentences), which can make it more difficult for the translator to retain a sense of the whole text. Although CAT tools can provide cognitive relief by making it easier to be consistent, they can also increase the load by making it harder for the translator to maintain cohesion and coherence (**Krüger2016**). Another challenge relates to the phenomenon of priming, in which "cognitive facilitation [...] is triggered by linguistic overlap between earlier and current processing" (**VandepitteEtAl2018**: 362). For example, suggestions that are not suitable can serve as inspirations for translators to produce versions that are (**Farrell2023**). However, the sheer presence of a TM match or MT suggestion can also block cognitive processing and force the translator to exert extra effort to mitigate the effects of priming (**Kolb2024**).

3.1.3 Drafting translations with MT or GenAI

A translation process that includes producing a draft with an MT engine or GenAI is different from translating with a CAT tool or from scratch in certain ways, yet quite similar in others. The similarities lie in evaluating the translation solutions through a careful bilingual review and adapting the TT to be more appropriate. In all types of translation processes, thoroughly understanding the ST and recognising errors, misrepresentations and unmotivated omissions or additions in the draft are just as important as being able to edit the latter to produce a fluent, cohesive and coherent TT.

Translation processes that incorporate GenAI shift the focus from writing to reading, prompting, evaluating and editing. Using a GenAI system with a simple prompt to translate a ST can be expected to produce a draft with features typical of MT output, the quality of which will depend on the particular engine being accessed. The process of improving such output through PE differs in many respects from editing or revision (do **CarmoMoorkens2021**). As well as a careful monolingual reading of the output, PE should always involve a bilingual review to eliminate any bias and inappropriate additions, omissions or "hallucinations" randomly introduced by the engine (see **DaleEtAl2023**).

Working with GenAI in a more informed way necessitates a prompting process that provides detailed information about the purpose of the translation, client guidelines, target genre and audience. An iterative process of prompting, evaluation and reprompting to obtain suitable output could precede a prompting sequence to eliminate any problems such as terminological inconsistencies and unnecessary repetition. Similar issues to those mentioned above for MT output (i.e. bias, inappropriate additions and omissions, hallucinations) also need to be rectified.

3.2 Describing translation skills in the GenAI era

It has been convincingly argued elsewhere (e.g. **NitzkeHansen-Schirra2021**; O'**Brien2021**) that using MT to produce high-quality TTs requires a special set of skills, and those arguments also hold when GenAI is simply used to produce a translation of a ST. However, informed use of GenAI includes the awareness that the pre- and drafting phase of creating a TT requires well-considered prompting, and post-drafting demands careful editing. In the following, we consider textual and digital skills that should be foregrounded when training students to work with GenAI to produce TTs as well as the interlingual competence that needs to be developed for this.

3.2.1 3.2.1 Textual skills

As **Lafeber2023** has reported, professional translators should be able to "acquire subject-matter knowledge quickly [...] understand complex topics, figure out obscure meaning, appreciate the authors' intentions and the readers' needs". This high level of literacy suggests that a solid foundation in text analysis for translation (e.g. **Nord2005**) is needed as well as the ability to recognise one's own gaps in knowledge. Rather than hoping that an MT engine or GenAI will produce output that is easier to understand, students should be trained to identify difficult ST passages and to do the research needed to make sense of them. **GimpelEtAl2023** point out that "[l]earners must possess adequate knowledge of the subject under scrutiny to achieve satisfactory outcomes". In the context of translation, this means that students must have the skill to quickly search for resources (e.g. reference words, parallel texts, videos, images) to help them understand the ST. They also need to have efficient, purposeful reading skills to evaluate the usefulness of the respective resource for their own comprehension and its potential as a GenAI prompt.

Since GenAI can produce fluent, grammatically correct texts, the fostering of text production skills should focus on revising, editing and proofreading (see also **KoponenEtAl2021**). This represents a significant shift from the teaching of foreign languages and basic writing skills work that is still done in many entry-level degree programmes, especially for translation into the L2 (Cerezo **HerreroEtAl2021**) and consistent with recent proposals for curriculum development (e.g. **SawyerEtAl2019**).

3.2.2 3.2.2 Technological and digital skills

The technology competence described in the **EMT2022** framework potentially covers GenAI in stating that "students know how to use the most relevant IT applications [...] and adapt rapidly to new tools and IT resources". The need for a basic understanding of MT and data literacy are also referred to, which are subsumed under the concept of MT literacy in the context of translation (see Bowker and Buitrago **Ciro2019**). However, the EMT descriptions seriously underspecify what type of knowledge is needed and which skills should be fostered to achieve competence when working with GenAI.

Scholars outside Translation Studies have started arguing that all students need to acquire GenAI literacy. For example, **Pretorius2023** points out risks and challenges such as inaccurate results, unequal access to the technology, bias propagation and ethical concerns about sensitive data. We would add that students need to understand the consequences of the data sources used for the LLMs and the time lags associated with the latter's knowledge updates, which can be in the order of months if not years.

Perhaps the most noticeable change in technological and digital skills in an age of GenAI concerns the critical thinking required to formulate effective prompts, evaluate intermediate versions and adjust subsequent prompts accordingly. This also includes the ability to recognise the diminishing returns of continued prompting and the need for use one's own textual skills to polish the TT. Working with GenAI is different from simply using language technology in that the intermediate versions can be considered a type of priming or mutual prompting between the conscious student/translator and the non-conscious AI agent. Only continuous critical reflection can counteract potentially overtrusting GenAI output during the translation process. Technological and digital translation skills in an age of GenAI might therefore be more broadly conceptualised as key components of an HMI/HAI competence capable of accounting for the particular reciprocities of human and AI agency.

3.2.3 Interlingual competence

Prerequisites for professional translation have always included high reading proficiency in the SL as well as familiarity with the source culture and domain. No matter which aids are being used, the translator also needs high proficiency in the TL – both reading and revision skills – as well as in-depth knowledge of the domain and target culture. Interlingual transfer might be called for less often on the part of the professional translator deploying MT or GenAI, although they are always engaged in interlingual action.

As **Lafaber2023** found in her survey of institutional translators, professionals are expected to "achieve high levels of accuracy in their translations, conveying not only nuances but also intended effect [...] draft well in their TL, compensating for poor wording in the original when appropriate while adhering to in-house conventions". Just as actuaries attain high degrees of numeracy during their training but rarely do calculations by hand in their jobs, students should be given ample opportunity to translate with and without aids. The resulting interlingual competence in combination with HMI/HAI competence will contribute to their translation literacy (see also **Massey2021**), effectively preparing them to work in the language industry in an age of GenAI.

3.3 Emerging roles in the language industry

Translation training institutions are assumed to prepare students for the language industry, yet up to two-thirds of graduates may not actually work as translators (**HaoPym2023**: 223). However, according to the same authors, many of them work in language-related professions. This suggests that programmes would do well to also prepare their students for jobs without "translator" in the title. Even when translators are explicitly being recruited, though, a broad range of skills seems to be expected. In their analysis of job notices from 2005-2020, Prieto Ramos and **Guzmán2023** found that the duties of translators at supranational and intergovernmental organisations also included (in order of overall average mention): assistance with other tasks, terminology work, revision, CAT-tool management and editing.

On the topic of how automation is changing the translation profession, **PymTorres-Simón2021** discuss various recommendations for translators to focus on what machines cannot (yet) do, such as language service advice, service provision, language consulting and high-stakes communication. This is consistent with observations from industry observers that text production with GenAI is less suitable for "content that has a regulatory or technical purpose" (**Slator2024**). The same report also predicts increased demand for multilingual experts to revise corporate-generated output to meet expected quality levels (**Slator2024**).

LSPs already offer a wide range of AI-related services (**Slator2024**), many of which require linguistic expertise that may still be difficult to find (cf. **FaesEtAl2024**). Although only about 10% of language companies currently report that they use GenAI, about 40% planned to do so (**ELIS2024**). This represents an opportunity for both training institutions and their graduates and, encouragingly, over 60% of the students participating in the same survey reported that GenAI topics were part of their training (**ELIS2024**).

This raises one last important question about those who train translation students. The teachers themselves patently require the (Gen)AI literacy needed to understand AI techniques, critically assess AI productions and recommendations, and use AI creatively in their teaching (**OECD2023**).[9] And institutions need to have the staff development procedures in place that empower their teachers to do so.

4 Final take-aways

Whether employed as translators, linguists, transcreators, localisers, consultants or language professionals, our graduates have an important role to play in the language industry despite or because of the introduction of GenAI. But they will need appropriate skills to do so. And those skills will, above all, have to accommodate the dynamic interactions of human and AI agency. Though degrees of domain specialisation are open to debate and will need to be determined by local conditions, current TC models already adequately cover the old-school translation and textual skills that industry demands. But their transferable and technological/digital skills are underspecified, and the introduction of GenAI calls for greater precision.

We recommend that collaborative experiential learning should always integrate decisions about whether and how to deploy GenAI, together with critical evaluations of the resultant processes and products. Outcomes should focus on developing specific transferable skills:

- critical thinking
- adaptivity
- creative problem-solving
- cognitive and emotional self-regulation
- self-efficacy
- accountability and a sense of ethics
- collaborative ability in human and AI interactions.

[9] Ideas for course design, materials and teaching scenarios that align well with the skills and competences outlined here can be found in **PymHao2025**.

The extension of the last point aims to develop what we have termed HMI/HAI competence, which goes beyond the technological/digital and MT literacies described in existing TC models to include:

- engineering effective prompts (translation purpose, client specifications, target genre, audience, etc.), evaluating GenAI responses with supplementary research (where appropriate) and reprompting
- recognising priming effects and mitigating negative ones
- identifying and eliminating bias, additions, omissions, hallucinations
- understanding LLM data sources and knowledge time-lags.

None of the above should be regarded as optional add-ons but must receive at least equal weight to so-called core skills. Only then will our graduates be properly equipped for the GenAI age, whatever roles they take on.